In [1]:
```
pip install_seaborn
```

```
Note: you may need to restart the kernel to use updated packages.
ERROR: unknown command "install_seaborn" - maybe you meant "install"
```

In [2]:
```
import warnings
warnings.filterwarnings("ignore",category=FutureWarning)
```

In [1]:
```
import seaborn as sns
sns.get_dataset_names()
```

Out[1]:
```
['anagrams',
 'anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
 'exercise',
 'flights',
 'fmri',
 'geyser',
 'glue',
 'healthexp',
 'iris',
 'mpg',
 'penguins',
 'planets',
 'seaice',
 'taxis',
 'tips',
 'titanic']
```

In [3]:
```
import seaborn as sns
import matplotlib.pyplot as plt

#Load sample dataset
tips=sns.load_dataset("tips")

#Set a visual theme
sns.set_theme(style="darkgrid")
```

In [4]:
```
tips
```

Out[4]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

In [10]:
```
#Save to CSV
tips.to_csv("tips_dataset.csv",index=False)
```

In [11]:
```
import os
os.getcwd()
```

Out[11]:
```
'C:\\Users\\Aisharani Patro\\VS_CODE_PROJECT'
```

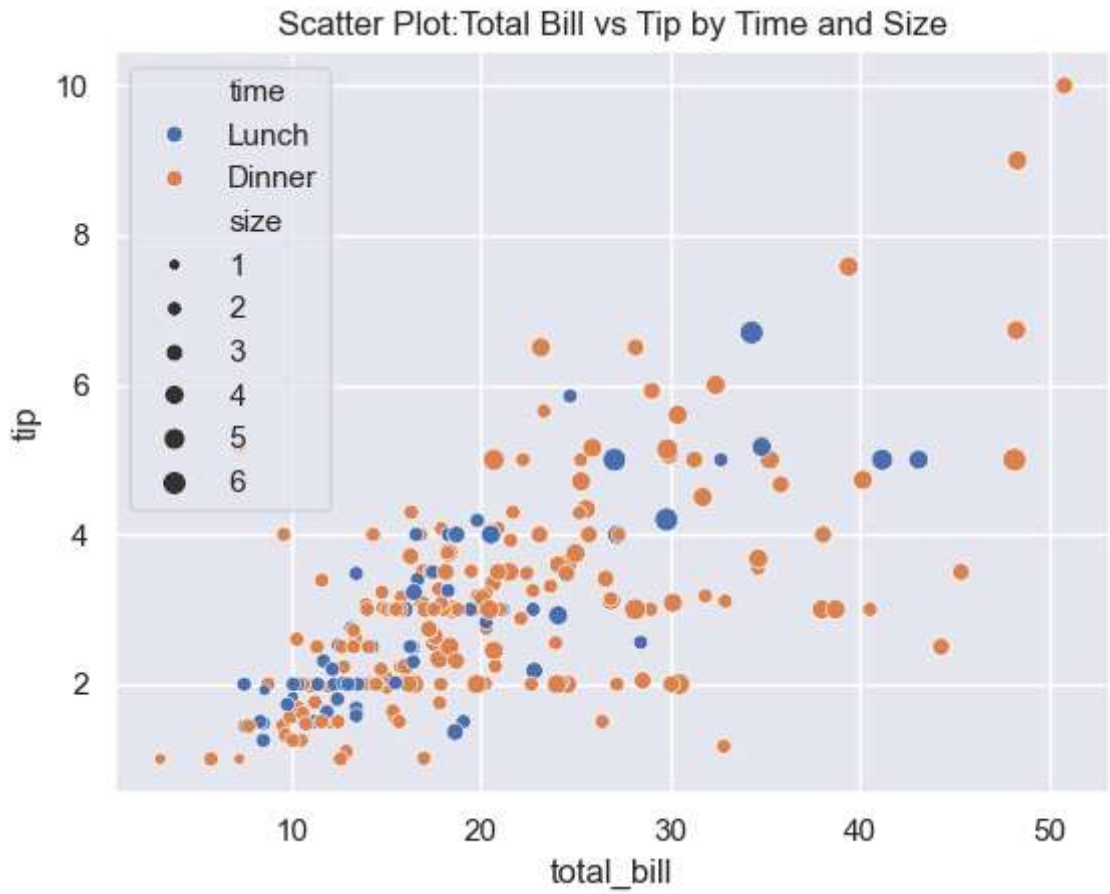
In [12]:
```
plt.figure(figsize=(8,6))
```

Out[12]:
```
<Figure size 800x600 with 0 Axes>
```

```
<Figure size 800x600 with 0 Axes>
```

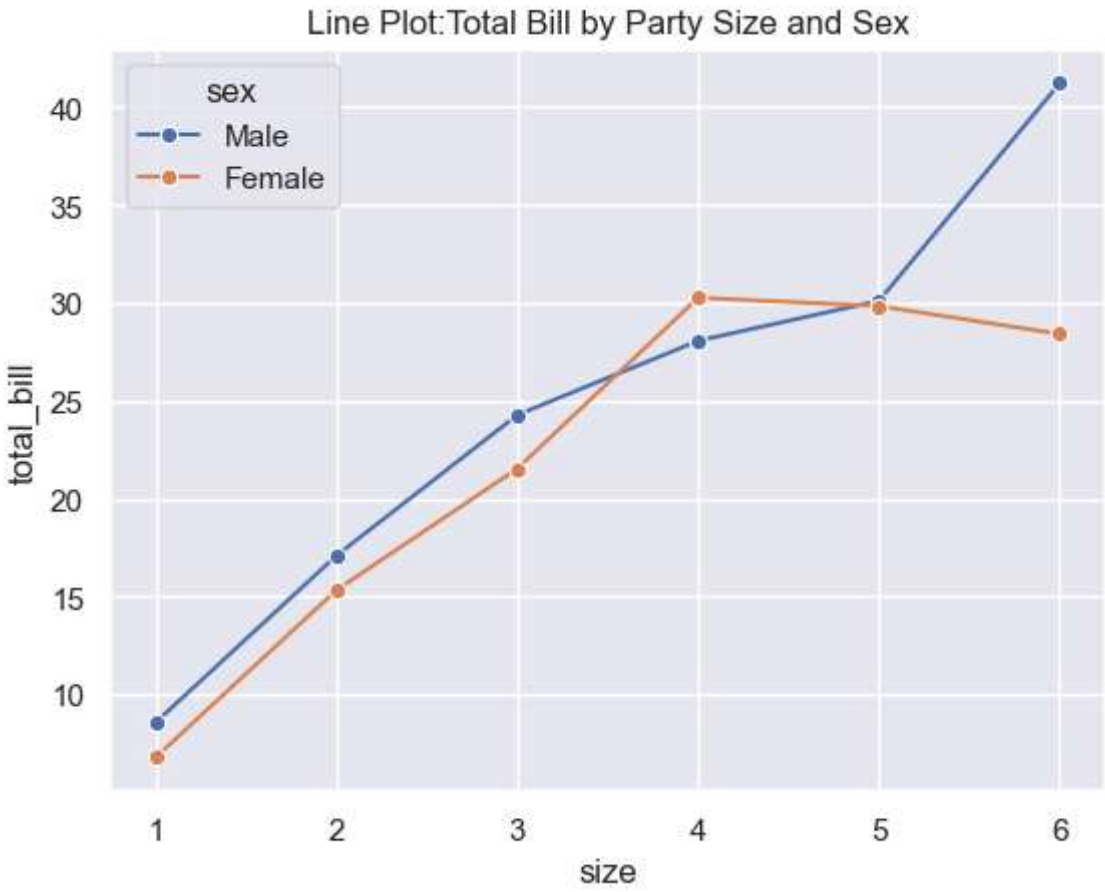
1.Scatter Plot

In [13]:
```
#1.Scatter Plot:Total Bill vs Tip,with hue and size

sns.scatterplot(data=tips,x="total_bill",y="tip",hue="time",size="size",palette="de
plt.title("Scatter Plot:Total Bill vs Tip by Time and Size")
plt.show()
```



2.Line Plot

In [14]:
```
#2.Line Plot: Total Bill by party size,split by sex

sns.lineplot(data=tips,x="size",y="total_bill",hue="sex",ci=None,marker="o")
plt.title("Line Plot:Total Bill by Party Size and Sex")
plt.show()
```

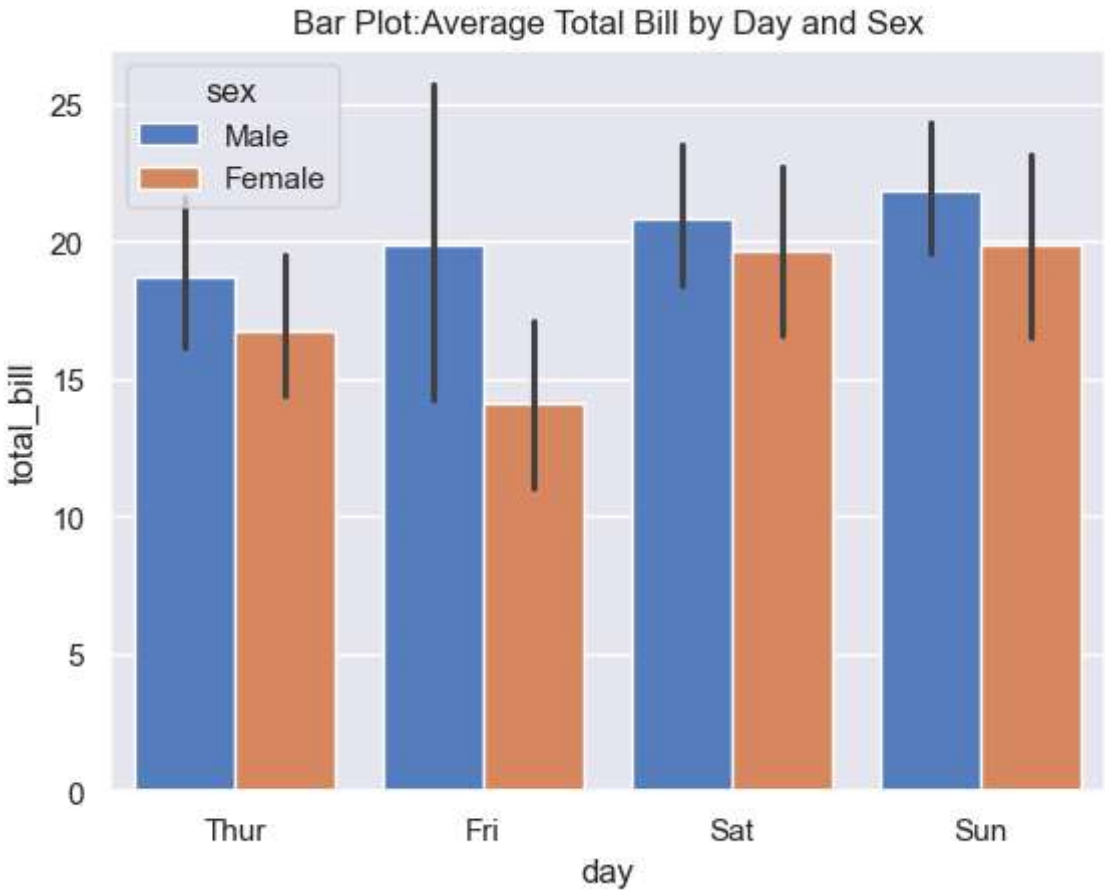


In [15]:
```
tips.columns
```

Out[15]:
```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='objec
t')
```

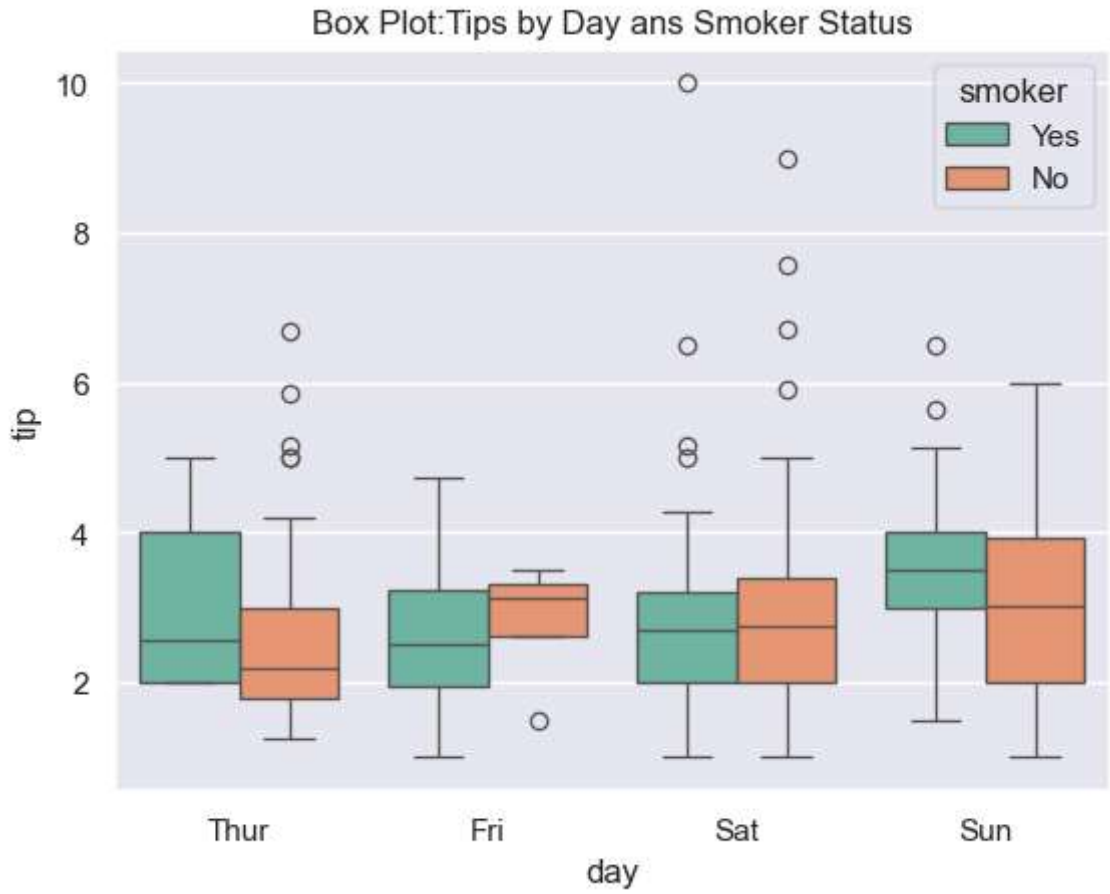
3.Bar Plot

In [16]:
```
#3.Bar Plot:Average Total Bill by Day,split by Sex

sns.barplot(data=tips,x="day",y="total_bill",hue="sex",palette="muted")
plt.title("Bar Plot:Average Total Bill by Day and Sex")
plt.show()
```



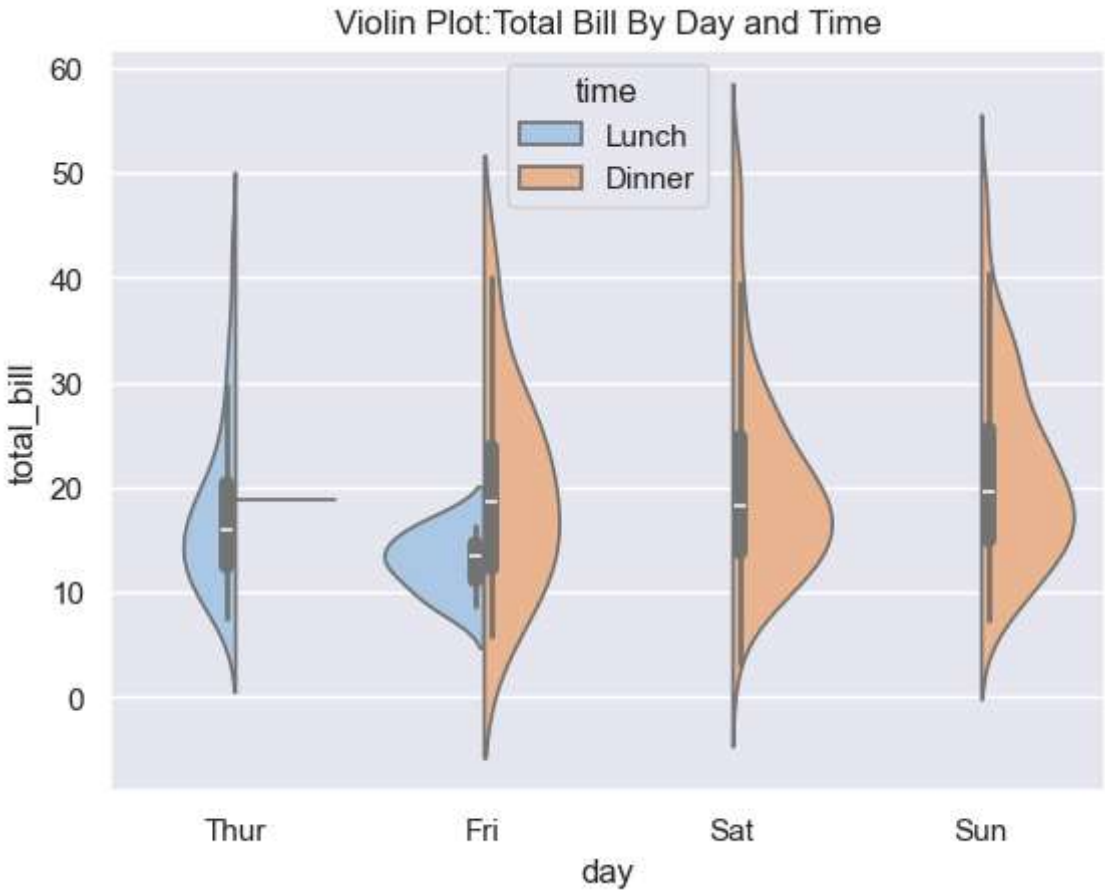
4.Box Plot

In [17]:
```
#4.Box Plot:Total Bill by Day,split by Time

sns.boxplot(data=tips,x="day",y="tip",hue="smoker",palette="Set2")
plt.title("Box Plot:Tips by Day ans Smoker Status")
plt.show()
```



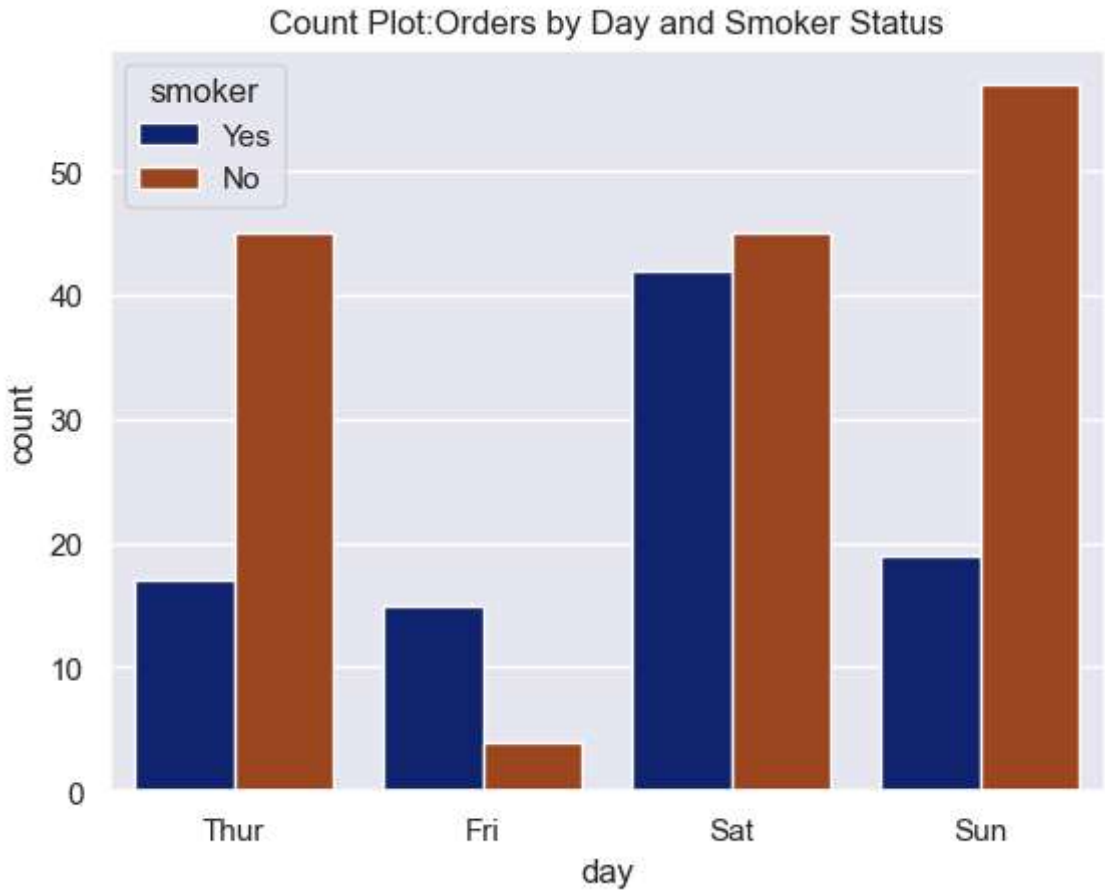
5.Violin Plot

In [18]:
```
#5.Violin Plot:Total Bill by Day,split by Time

sns.violinplot(data=tips,x="day",y="total_bill",hue="time",split=True,palette="past
plt.title("Violin Plot:Total Bill By Day and Time")
plt.show()
```



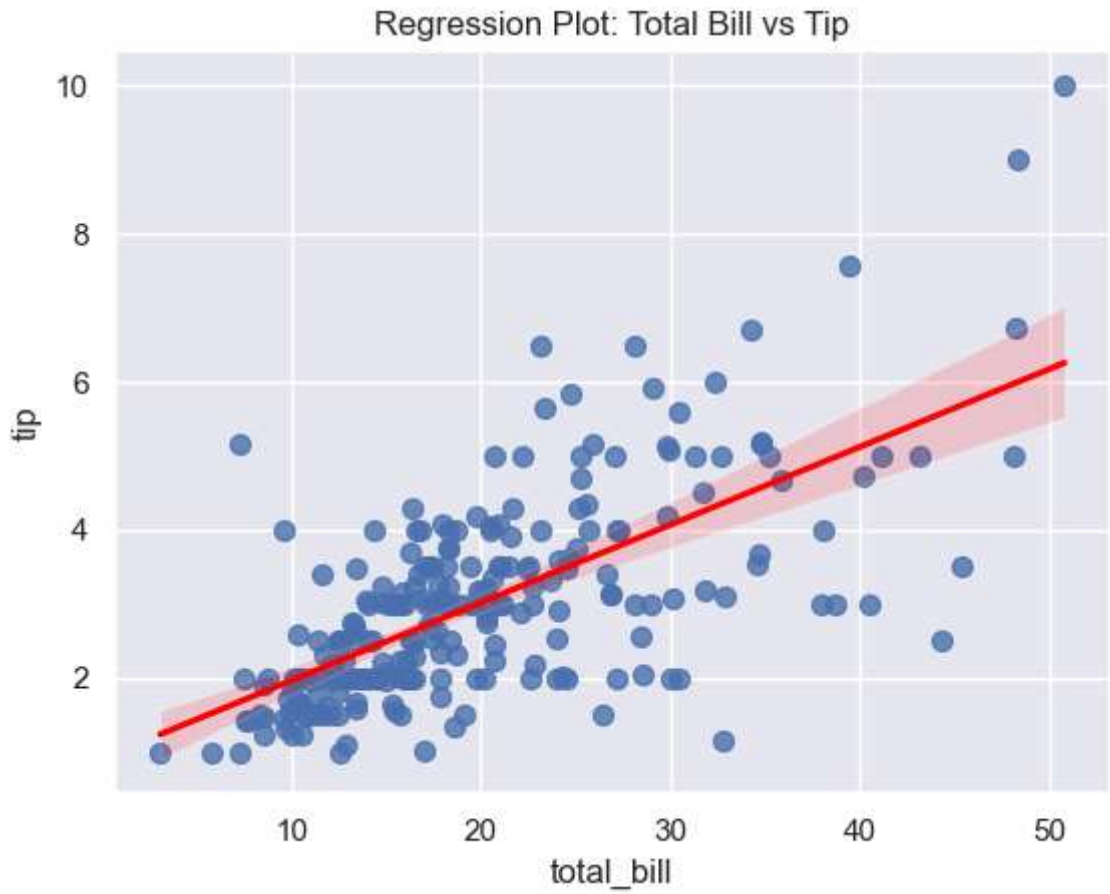
6.Count Plot

In [19]:
```
#6.Count Plot:Orders by Day,split by Smoker
sns.countplot(data=tips,x="day",hue="smoker",palette="dark")
plt.title("Count Plot:Orders by Day and Smoker Status")
plt.show()
```



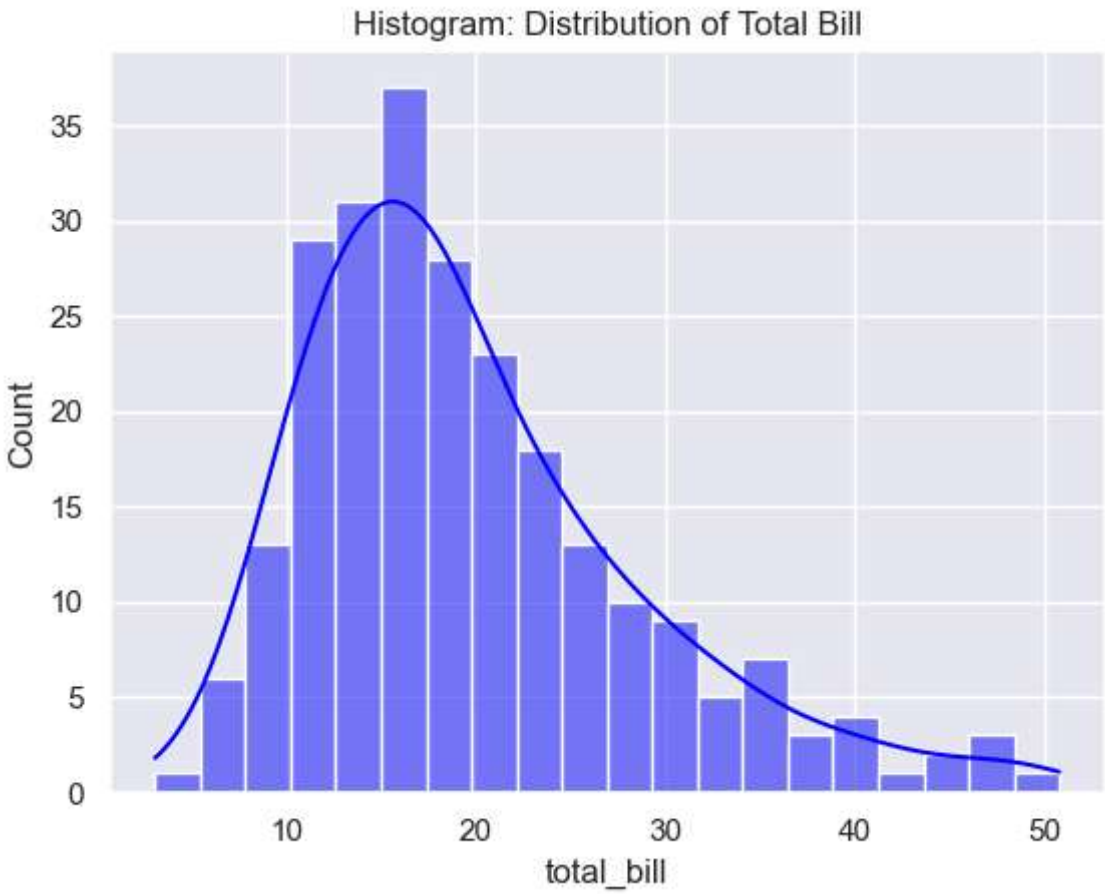
7.Regression Plot

In [20]:
```
# 7. Regression Plot: Total Bill vs Tip with regression line

sns.regplot(data=tips, x="total_bill", y="tip", scatter_kws={"s": 50}, line_kws={"c
plt.title("Regression Plot: Total Bill vs Tip")
plt.show()
```



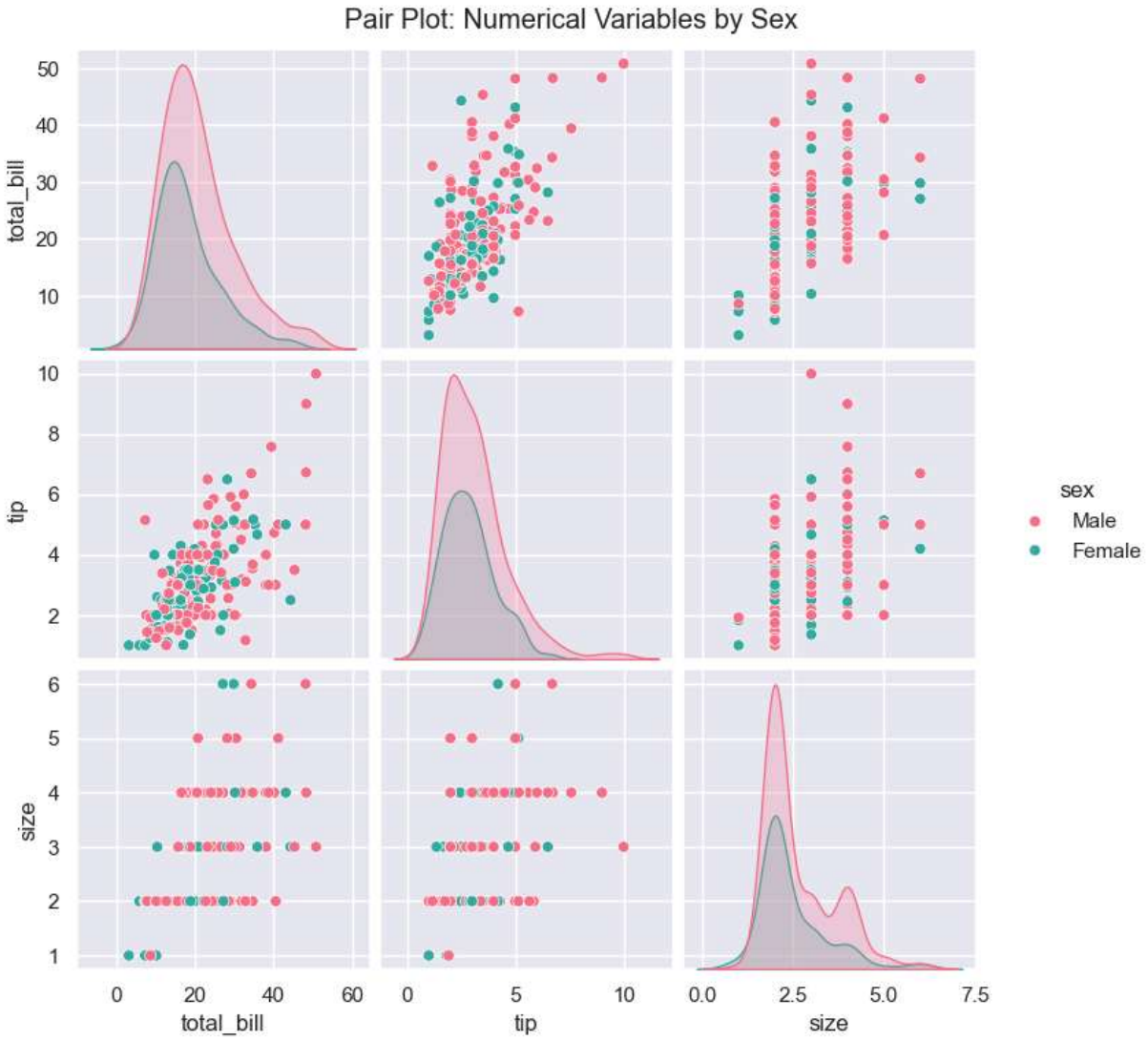
8.Histogram

In [21]:
```
#8. Histogram (Distribution Plot): Total Bill with KDE

sns.histplot(data=tips, x="total_bill", kde=True, bins=20, color="blue")
plt.title("Histogram: Distribution of Total Bill")
plt.show()
```



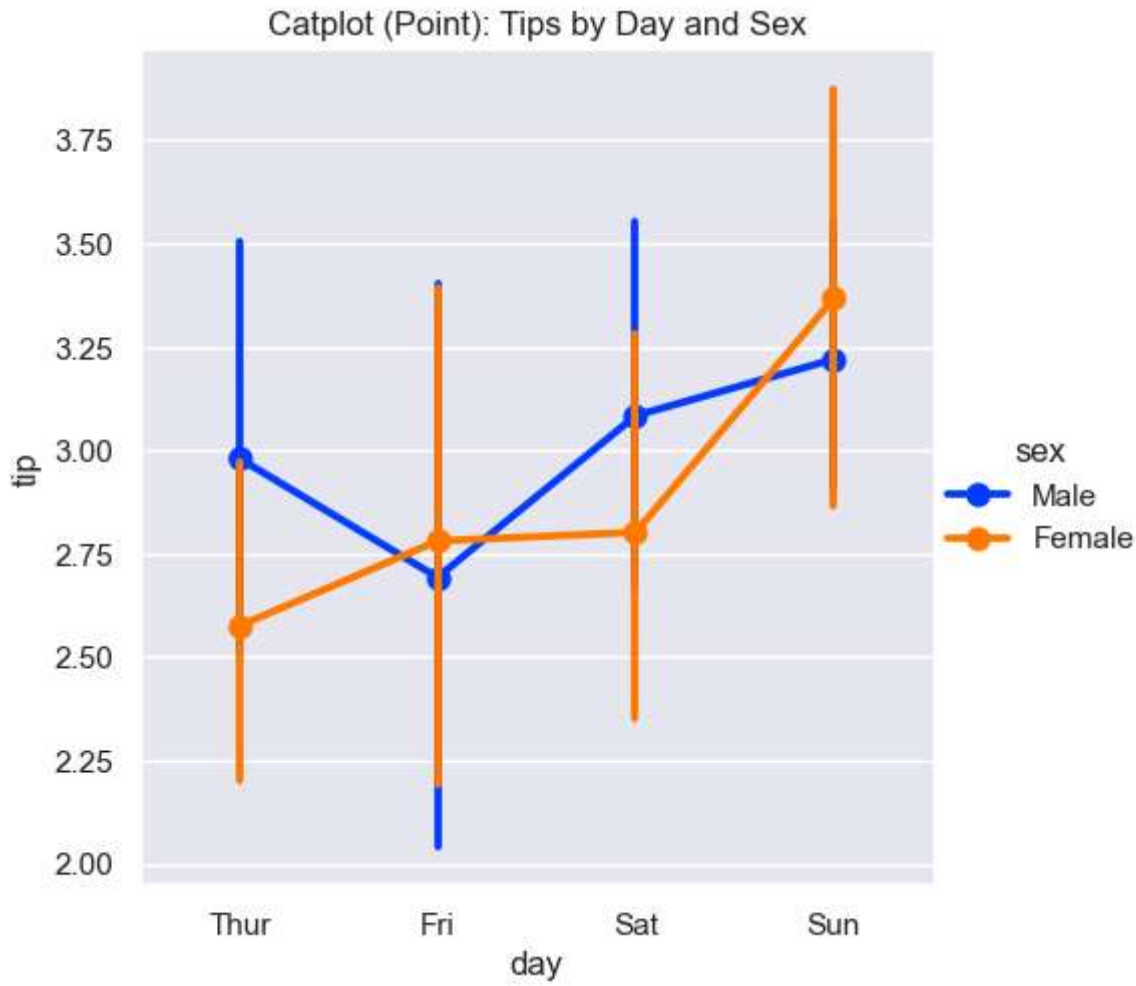
9.Pair Plot

In [22]:
```
#9. Pair Plot: Relationships between numerical variables

sns.pairplot(tips, hue="sex", vars=["total_bill", "tip", "size"], palette="husl")
plt.suptitle("Pair Plot: Numerical Variables by Sex", y=1.02)
plt.show()
```



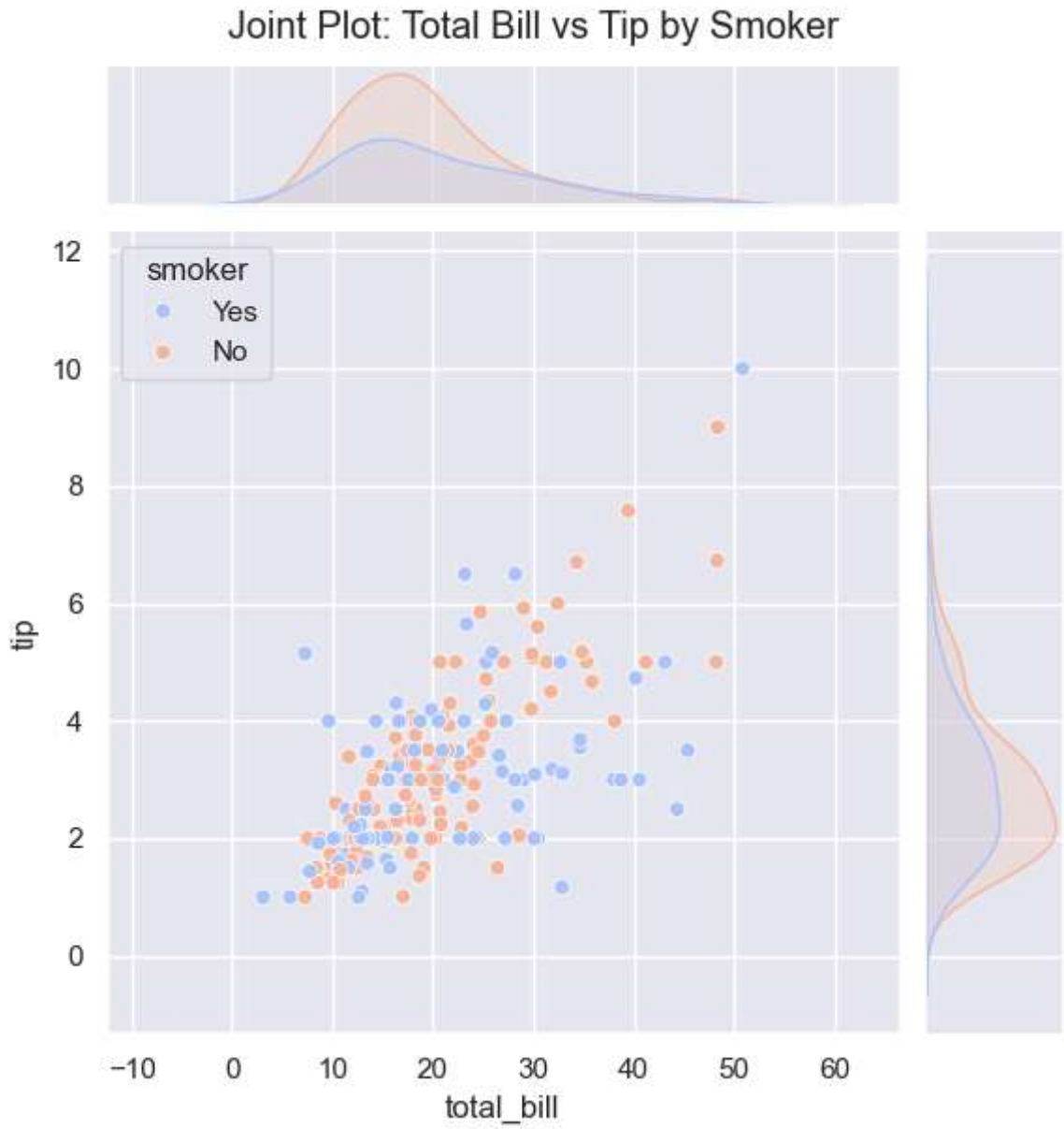
10.Catplot

In [23]:
```
# 10. Catplot (Point Plot): Tip by Day and Sex

sns.catplot(data=tips, x="day", y="tip", hue="sex", kind="point", palette="bright")
plt.title("Catplot (Point): Tips by Day and Sex")
plt.show()
```



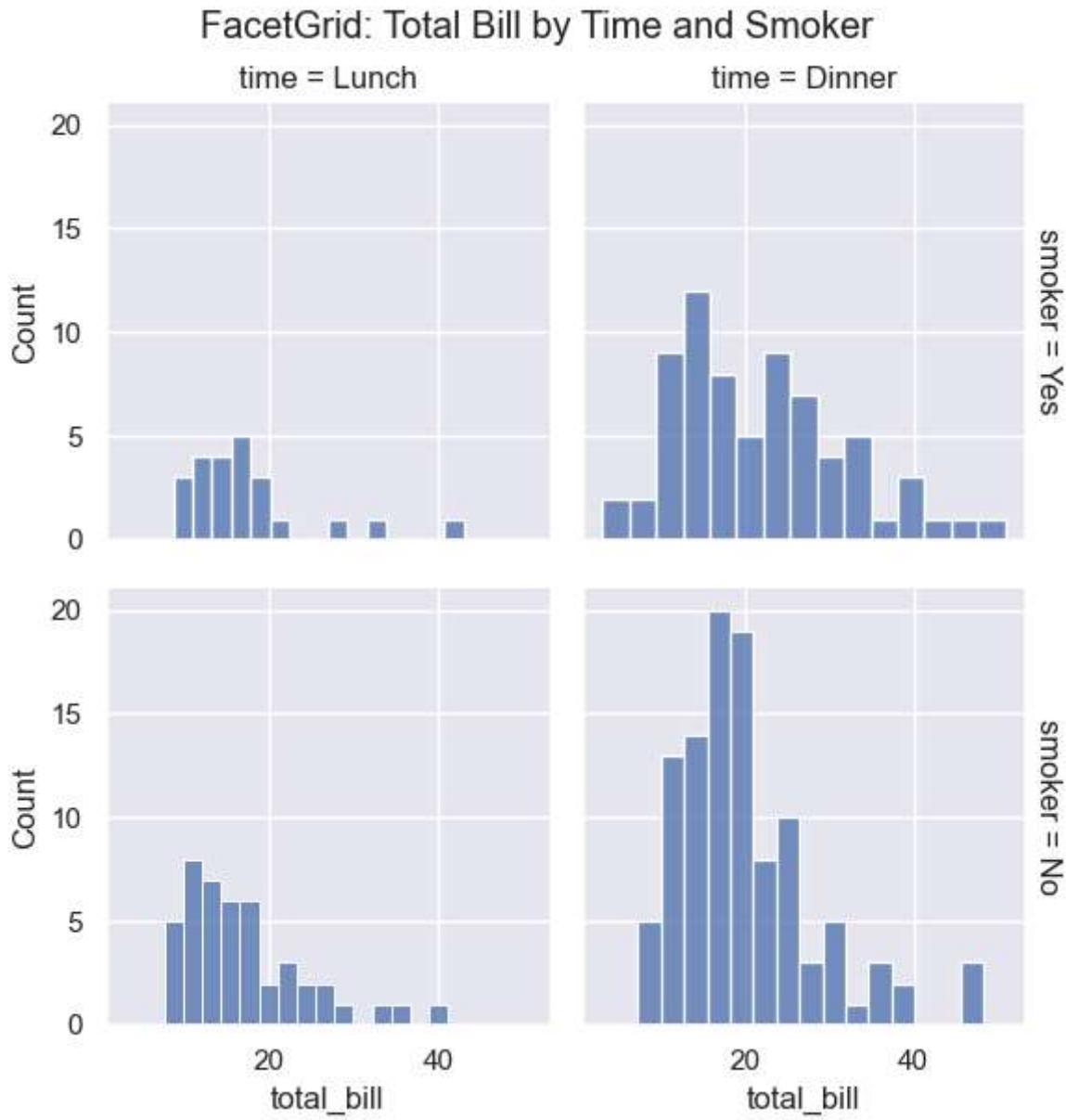
11.Joint Plot

In [24]:
```
# 11. Joint Plot: Total Bill vs Tip with marginal distributions
sns.jointplot(data=tips, x="total_bill", y="tip", kind="scatter", hue="smoker", pal
plt.suptitle("Joint Plot: Total Bill vs Tip by Smoker", y=1.02)
plt.show()
```



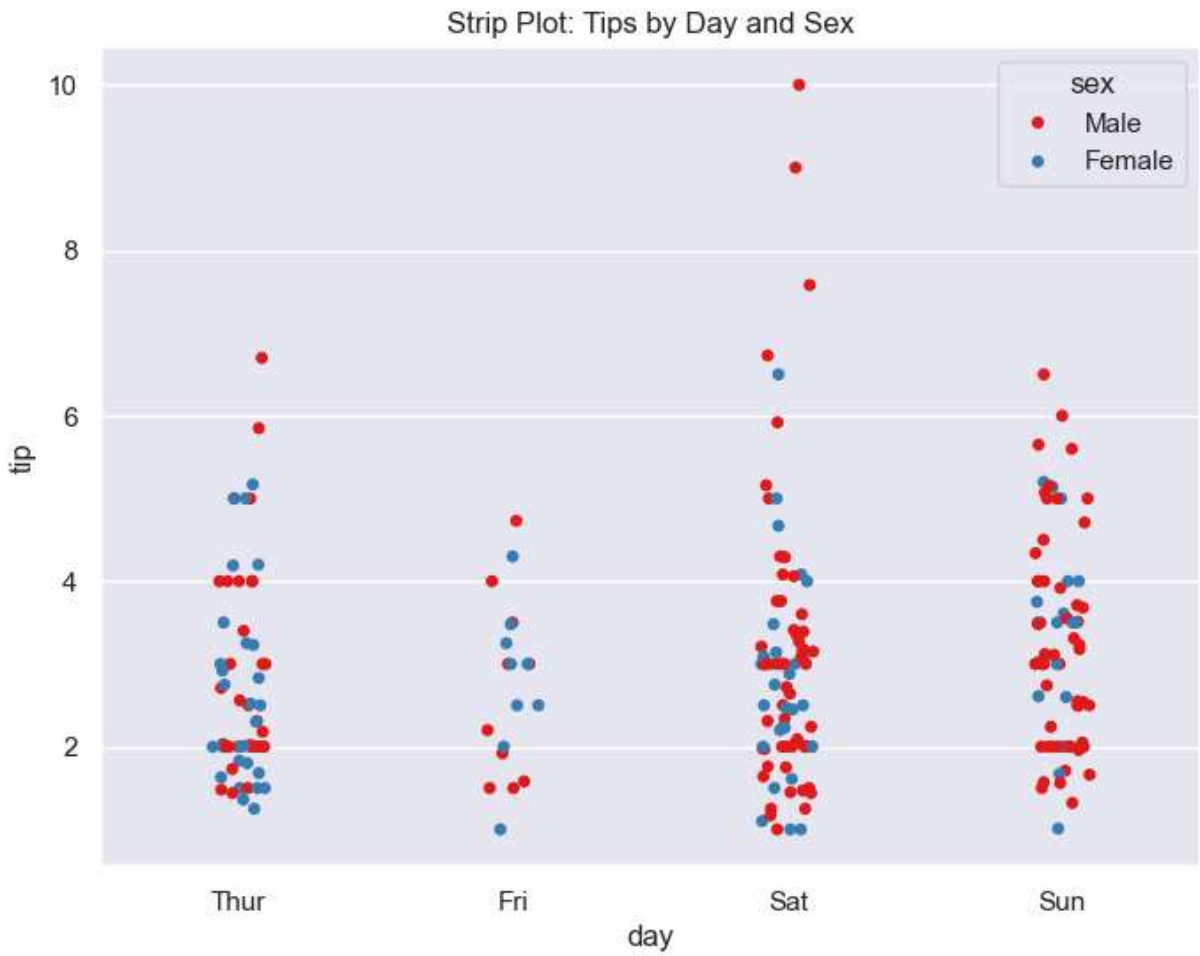
12.FacetGrid

In [25]:
```
# 12. FacetGrid: Total Bill by Day, faceted by Time and Smoker
g = sns.FacetGrid(tips, col="time", row="smoker", margin_titles=True)
g.map(sns.histplot, "total_bill", bins=15)
g.fig.suptitle("FacetGrid: Total Bill by Time and Smoker", y=1.02)
plt.show()
```



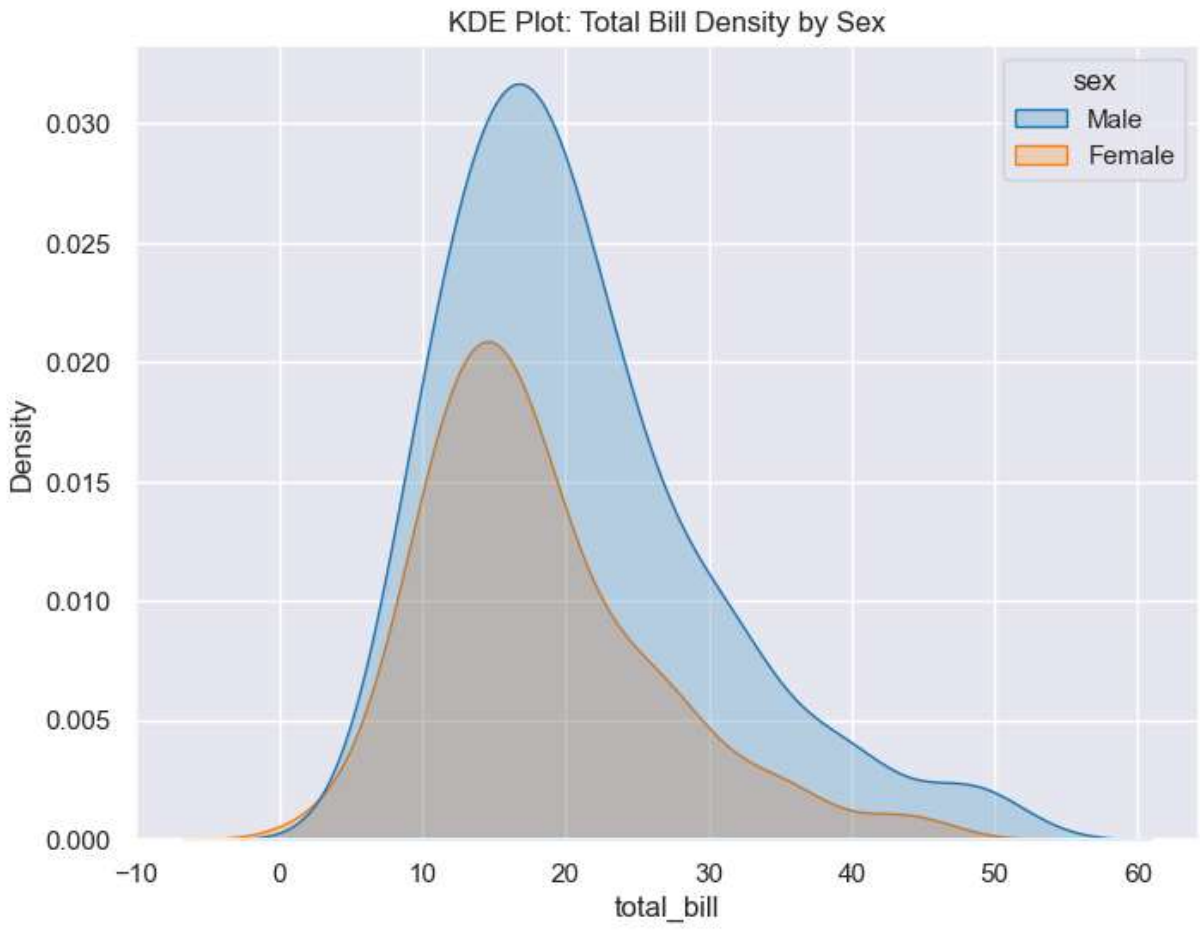
13.Strip Plot

In [26]:
```
# 13. Strip Plot: Tips by Day, colored by Sex
plt.figure(figsize=(8, 6))
sns.stripplot(data=tips, x="day", y="tip", hue="sex", palette="Set1", jitter=True)
plt.title("Strip Plot: Tips by Day and Sex")
plt.show()
```



14.KDE Plot

In [27]:
```
# 14. KDE Plot: Total Bill density by Sex
plt.figure(figsize=(8, 6))
sns.kdeplot(data=tips, x="total_bill", hue="sex", fill=True, palette="tab10")
plt.title("KDE Plot: Total Bill Density by Sex")
plt.show()
```


KDE Plot: Total Bill Density by Sex
sex
Male
Female
Density
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−10
0
10
20
30
40
50
60
total_bill